



NATIONAL SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2023

MATHEMATICAL LITERACY P1 MARKING GUIDELINE

MARKS: 100

Symbol	Explanation
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG/RM	Reading from a table/Reading from a graph/Reading from a map
F	Choosing the correct formula
SF	Substitution in a formula
J	Justification
P	Penalty, e.g., for no units, incorrect rounding off etc.
R	Rounding Off/Reason
AO	Answer only
NPR	No penalty for correct rounding

This marking guideline consists of 6 pages.

QUESTION 1 [23 MARKS]			
Ques.	Solution	Explanation AO: FULL MARKS	T&L
1.1.1	$A = R33\,150 + R1\,625 \checkmark MA$ $= R34\,775 \checkmark A$	1MA for adding correct values 1A answer (2)	F L1
1.1.2	$R33\,150 \div 26 \checkmark MA$ $= R1\,275,00 \checkmark A$	1MA for dividing correct values 1A answer (2)	F L1
1.1.3	$B = R7\,927,86 - (R375 + R850,25 + R1\,500) \checkmark MA$ $= R7\,927,86 - R2\,725,25$ $= R5\,202,61 \checkmark A$	1MA for subtracting correct values 1A answer (2)	F L1
1.1.4	$\checkmark RT$ Percentage = $\frac{R850,25}{R7\,927,86} \times 100 \checkmark M$ $= 10,72\% \checkmark CA$	1RT correct values 1M multiplying by 100 1CA answer (3)	F L1
1.2.1	$R22,50 \checkmark \checkmark RT$	2RT reading correct value (2)	F L1
1.2.2	Vaal Plaza Main line $\checkmark \checkmark RT$	2RT correct tollgate (2)	F L1
1.2.3	$140 : 225 \checkmark RT$ $28 : 45 \checkmark A$	1RT correct values 1A simplifying correctly (2)	F L1
1.2.4	$R271,00 \times 2 \checkmark MA$ $= R542,00 \checkmark A$	1MA multiplying correct rate by 2 1A answer (2)	F L1
1.3.1	Pie Chart $\checkmark \checkmark RT$	2RT correct graph (2)	D L1
1.3.2	$\checkmark MA$ Percentage comedies = $100\% - (21\% + 6\% + 24\% + 11\%)$ $= 38\% \checkmark A$	1MA subtracting 1A percentage (2)	D L1
1.3.3	Number of romance = $24\% \times 220 \checkmark MA$ $= 52,8$ $= 53 \text{ students } \checkmark A$	1MA multiplying correct percentage 1A correct answer Accept 52 (2)	D L1
		[23]	

QUESTION 2 [25 MARKS]			
Ques	Solution	Explanation/Marks AO: FULL MARKS	T/L
2.1.1	$\begin{aligned} &\checkmark M \quad \checkmark RT \\ \text{Jan 2023: } R12,47 \times 1,069 &= R13,33 \quad \checkmark A \\ &\checkmark MA \\ \text{Feb 2023: } R13,33 \times 1,07 &= R14,26 \quad \checkmark CA \end{aligned}$	1RT correct rate 1M multiplying 1A answer 1MA multiplying with correct rate 1CA answer (5)	F L3
2.1.2	Prices increased $\checkmark A$ Prices still went up, but by a lower percentages/ the percentages are not negative. $\checkmark O$	1A increased 1O reason (2)	F L4
2.2.1	$\checkmark A \quad \checkmark A$ 7 May 2023	1A 7 th 1A May (2)	F L1
2.2.2	$\begin{aligned} \text{Total} &= R349 \times 3 \quad \checkmark MA \\ &= R1\,047 \quad \checkmark A \\ &\approx R1\,000 \quad \checkmark R \end{aligned}$	1MA multiplying by 3 1A answer 1R rounding to nearest 1 000 (3)	F L2
2.2.3	$\begin{aligned} \text{Labour per hour} &= R6\,500 \div 7,5 \text{ hrs} \quad \checkmark MA \\ &= R866,67 \quad \checkmark A \end{aligned}$	1MA dividing by 7,5 1A correct answer (2)	F L2
2.2.4	$\begin{aligned} \text{Total} &= R5\,499 + R1\,047 + R1\,699 + R6\,500 \quad \checkmark M \\ &= R14\,745 \times 1,15 \quad \checkmark MA \\ &= R16\,956,75 \quad \checkmark CA \end{aligned}$	CA from 2.2.2 1M adding values 1MA multiplying by 1,15 1CA answer (3)	F L2
2.2.5	$\begin{aligned} \% \text{ change} &= \frac{5\,795 - 5\,499}{5\,499} \times 100\% \quad \checkmark MA \\ &= 5,38\% \quad \checkmark CA \\ \therefore \text{Valid} &\quad \checkmark O \end{aligned}$	1RT correct values 1MA correct % calculation 1CA simplification 1O statement (4)	F L4
2.3	$\begin{aligned} \text{Block 1: } 600 \text{ kWh} \times 229,00c &= 137\,400c \quad \checkmark MA \\ \text{Block 2: } 112 \text{ kWh} \times 278,46c &= 31\,187,52c \quad \checkmark MA \\ \text{Total} &= 168\,587,52c \div 100 \quad \checkmark C \\ &= R1\,685,88 \quad \checkmark CA \end{aligned}$	1MA multiplying Block 1 1MA answer Block 2 1C divide by 100 1CA answer (4)	F L3
		[25]	

QUESTION 3 [19 MARKS]			
Ques	Solution	Explanation	T&L
3.1	Forty-five million eight hundred thousand dollars ✓✓RT	2RT correct value (2)	D L1
3.2	Median: 5; 25; 47; 50; <u>52</u> ; 55; 55; 80; 90 ✓RT ✓MA = \$52 000 000 / \$52 million ✓A	1RT correct values 1MA method median 1A correct answer (3)	D L3
3.3	Federer total = \$121,2 – \$30,5 ✓M = \$90,7 million ✓A Total = \$130 + \$121,2 + \$115 + \$95 + \$92,8 + \$92,1 + \$90,7 + \$90 + \$83,9 ✓M = \$910,7 million / \$910 000 000 ✓CA	1M subtracting 1A simplification 1M adding 1CA simplification (4)	D L3
3.4	Mean = $\frac{\$451,7}{9}$ ✓M ✓MA = \$50,2 million / \$50 200 000 ✓CA	1M adding values 1MA dividing by 9 1CA answer (3)	D L2
3.5	Percentage = $\frac{55}{115} \times 100$ ✓MA = 47,83% ✓A	1MA correct values multiply by 100 1A simplification (2)	D L2
3.6	70 : 90 ✓RT 7 : 9 ✓CA	1RT correct values 1CA simplification of ratio (2)	D L2
3.7	Probability = $\frac{4}{9}$ ✓RT ✓RT = 0,444 ✓CA	1RT numerator 1RT denominator 1CA rounding to 3 decimal places (3)	P L2
		[19]	

QUESTION 4 [33 MARKS]			
Ques	Solution	Explanation/Marks	T&L
4.1.1	Dependent: Amount in Rands ✓RT Independent: Number of curry bunnies ✓RT	1RT dependent 1RT independent (2)	F L1
4.1.2	$R250 \div 10$ ✓MA $= R25,00$ ✓A	1MA dividing by 10 1A simplification (2)	F L1
4.1.3	Total Expenses = $450 + (10 \times \text{number of curry bunnies})$ ✓RT ✓RT	1RT fixed cost 1RT cost per curry bunny (2)	F L2
4.1.4	Income and Expenses of Curry Bunny Sales Amount in Rands Number of Curry Bunnies	1A starting point 1A 2 correct points 1A end point (3)	F L2
4.1.5	Profit ONE month = $R5\ 000 - R2\ 450$ $= R2\ 550$ ✓A Profit FOUR months = $R2\ 550 \times 4$ ✓MA $= R10\ 200$ ✓CA $\therefore$ Valid ✓O	1A profit one month 1MA profit 4 months 1CA answer 1O statement (4)	F L4
4.2.1	$A = 20 + 13$ ✓RT $= 33$ ✓A	1RT adding correct values 1A answer (2)	D L2
4.2.2	$15:00 - 17:59$ ✓✓A	2A correct answer (2)	D L2
4.2.3	$18:00 - 20:59$ ✓✓RT	2RT correct time interval (2)	D L2

Ques	Solution	Explanation/Marks	T&L
4.2.4	$6,5 \ell \times 77 = 500,5 \ell$ ✓MA Cost = $500,5 \ell \times R21,92$ ✓M $= R10\,970,96$ ✓CA	1MA calculating litres 1M multiplying 1CA answer (3)	D L3
4.2.5	Probabilty = $\frac{32}{77}$ ✓RT ✓RT	1RT numerator 1RT denominator (2)	P L2
4.3.1	BANK B ✓RT Interest amount is the same for each year. ✓O	1RT correct bank 1O statement/opinion (2)	F L2
4.3.2	BANK A: $R1\,277,29 \times 1,085$ ✓MA $= R1\,385,86$ ✓A Difference = $R1\,385,86 - R1\,340$ ✓M $= R45,86$ ∴ Valid ✓O	1MA multiplying by rate 1A simplification 1M difference 1O statement (4)	F L4
4.3.3	$68,49 \times 19,83$ ✓MA $= R1\,358,16$ ✓A ∴ Yes, she will have enough. ✓O	1MA multiplying by exchange rate 1A simplification 1O statement (3)	F L4
		[33]	
		TOTAL: 100	